

Numerical simulation of (T_2 , T_1) 2D NMR and fluid responses*

Tan Mao-Jin^{1,2}, Zou You-Long², Zhang Jin-Yan³, and Zhao Xin³

Abstract: One-dimensional nuclear magnetic resonance (1D NMR) logging technology is limited for fluid typing, while two-dimensional nuclear magnetic resonance (2D NMR) logging can provide more parameters including longitudinal relaxation time (T_1) and transverse relaxation time (T_2) relative to fluid types in porous media. Based on the 2D NMR relaxation mechanism in a gradient magnetic field, echo train simulation and 2D NMR inversion are discussed in detail. For 2D NMR inversion, a hybrid inversion method is proposed based on the damping least squares method (LSQR) and an improved truncated singular value decomposition (TSVD) algorithm. A series of spin echoes are first simulated with multiple waiting times (T_{WS}) in a gradient magnetic field for given fluid models and these synthesized echo trains are inverted by the hybrid method. The inversion results are consistent with given models. Moreover, the numerical simulation of various fluid models such as the gas-water, light oil-water, and vicious oil-water models were carried out with different echo spacings (T_{ES}) and T_{WS} by this hybrid method. Finally, the influences of different signal-to-noise ratios (SNRs) on inversion results in various fluid models are studied. The numerical simulations show that the hybrid method and optimized observation parameters are applicable to fluid typing of gas-water and oil-water models.

Keywords: Two-dimensional NMR (2D-NMR) logging, hybrid inversion method, transverse relaxation time (T_2), longitudinal relaxation time (T_1), fluid typing

Introduction

Nuclear magnetic resonance (NMR) logging is a unique logging method to distinguish movable and bound fluids in formations and is widely used in hydrocarbon identification (Coats et al., 1998; Xiao, 1998; Wang et al., 2001; Xiao, 2007). However, the 1D NMR logging technology can only measure transverse relaxation

time (T_2) of fluids in pores, so it is greatly limited in identification and quantitative evaluation of fluids such as oil, gas, and water in some reservoirs. That is because when oil, gas, and water coexist in local pores their signals sometimes overlap each other in the T_2 spectrum. To distinguish them from each other, the usual differential spectrum method (DSM), shifted spectrum method (SSM), and enhanced diffusion method (EDM) are difficult (Wang et al., 2001; Shao, 2003; Tan et al., 2006).

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1. Key Laboratory of Geo-detection (China University of Geosciences, Beijing), Ministry of Education, Beijing 100083, China.

2. School of Geophysics and Information Technology, China University of Geosciences, Beijing 100083, China.

3. Shengli Well Logging Co., SINOPEC, Dongying 257096, China.

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(T_2 , T_1) 2D NMR and fluid responses

Two-dimensional nuclear magnetic resonance (2D NMR) logging is newly developed and can provide a variety of parameters including transverse relaxation time (T_2), observed longitudinal relaxation time (T_1), fluid diffusion coefficient (D), internal magnetic field gradient (G), and other parameters. The current 1D NMR logging cannot meet 2D observation requirements, so it has to redesign its observation activations and corresponding parameters. The new NMR instruments such as MREx of Baker Atlas and MRX of Schlumberger can both be used for 2D NMR logging.

Hurlimann et al. (2002) and Sun and Dunn (2005a; 2005b) proposed a method to improve spin-echo pulse sequence (CPMG) using two windows and achieved 2D NMR logging based on relaxation- diffusion (T_2 , D) analysis, namely (T_2 , D) 2D NMR, which greatly enhanced the accuracy of fluid identification and saturation calculations. Xiao (1998, 2007) and Xie et al. (2005, 2009) have performed some 2D NMR experiments with man-made paramagnetic sandstone and water-saturated natural shale sandstone and have given two-dimensional distributions of transverse relaxation time (T_2) and internal magnetic field gradient (G) namely (T_2 , G) 2D NMR, which is significant for NMR measurement analysis of complex terrestrial sedimentary rock strata. Gu and Liu (2007) and Gu et al. (2009) have studied traditional singular value decomposition (SVD), improved the SVD algorithm, and have achieved a two-dimensional continuous spectrum for NMR inversion and discussed the impacts of signal-noise-ratio (SNR) on inversion results. Tan et al. (2011) used the truncated singular value decomposition (TSVD) for inversion of (T_2 , D) 2D NMR. The experiment was performed with different magnetic field gradients with different echo spacing groups using gas-water models with different gas saturation and the response characteristics of the different saturation models were analyzed. Furthermore, the effects of different SNRs on inversion results were studied.

In this paper, we introduce and study the basic principles and mathematical model of 2D NMR logging with multiple waiting times (T_w) based on the NMR relaxation mechanism under conditions of the gradient magnetic field. Then, the LSQR and TSVD algorithms are used to realize (T_2 , T_1) 2D NMR inversion. Combined with the characteristics of the two algorithms, a hybrid inversion method for 2D NMR inversion is proposed using the LSQR first and then the TSVD algorithm is used for a second inversion. Whose initial guess is from the LSQR result. In the experiment, the hybrid inversion is better than a single algorithm. Using

this new inversion method, the 2D spectrum inversion of different fluid models are implemented and the influence of different combinations of multiple T_w and different T_E are studied in detail. The influences of different echo spacing, different signal to noise ratio (SNR) on the inversion results are studied. All the research results provide an important guidance for the parameter design for 2D NMR logging.

Methods and principles

The purpose of NMR logging is to make porosity calculations, pore structure, and fluids identification in porous media. The new generation of NMR logging tools can measure a number of spin-echo trains with different waiting times (T_w). In porous media, we suppose that s groups of spin echo trains with different waiting times (T_w) were observed and the echo amplitude in each echo train is relative to both a decaying component, $e^{-T_{i,s}/T_{2j}}$, and polarizing factor, $1 - e^{-T_{w,s}/T_{1r}}$, so the NMR echo train signals based on the CPMG pulse sequence are discrete are discrete as the following equation (Sun et al., 2005):

$$b_{is} = \sum_{j=1}^m \sum_{r=1}^p f_{jr} (1 - e^{-T_{w,s}/T_{1r}}) e^{-T_{i,s}/T_{2j}} + \varepsilon_{is}, \quad (1)$$

where b_{is} is the i^{th} echo amplitude of the s^{th} echo-train with waiting time $T_{w,s}$, f_{jr} is the proton number with transverse relaxation time (T_{2j}) and longitudinal relaxation time (T_{1r}), ε_{is} is the i^{th} echo noise of the s^{th} echo-train with waiting time $T_{w,s}$.

Equation (1) can also be written as the matrix equation

$$b_{is} = A_{is,jr} f_{jr} + \varepsilon_{is}, \quad (2)$$

where $A_{is,jr} = (1 - e^{-T_{w,s}/T_{1r}}) e^{-T_{i,s}/T_{2j}}$, $i=1, \dots, N_{Es}$, N_{Es} is the number of echoes with waiting time $T_{w,s}$, $s=1, \dots, w$, $r=1, \dots, p$, $j=1, \dots, m$, and w is the designed number of waiting times, and p and m are the numbers of T_2 and T_1 components divided, respectively.

The linear equation (2) may be also written as

$$Ax = b. \quad (3)$$

The linear equation system (3) is solved by the inversion method and the proton number amplitude (f_{ij}) of transverse relaxation time (T_{2j}) and longitudinal relaxation time (T_{1r}) can be calculated and the 2D NMR spectrum is derived and correspondingly constructed and which can be used for fluid typing identification and evaluation.

Inversion algorithm tests

For 2D NMR spectrum inversion problems, matrix $A_{m \times n}$ is a large sparse kernel matrix. $A_{m \times n}$ is dependent on the NMR logging in-site observation activation, observation parameters, and the grid mesh. The final $A_{m \times n}$ is a serious pathological matrix. In this paper, several inversion algorithms are chosen for checking inversion results, which are verified by the relative error between the model and inversion results,

$$R_x = \frac{\|x_{\text{mod}} - x_{\text{inv}}\|}{\|x_{\text{mod}}\|}, \quad (4)$$

where, x_{mod} is the model parameter and x_{inv} is the inversion result.

The LSQR method was proposed by Paige and Sanders in 1982 and is used for solving least squares problems with the Lanczos iteration method. The LSQR method has the advantage of simplified sparse matrix calculations and it is suitable for solving large sparse matrix problems (Liu and Liu, 2006; Yang et al., 2008).

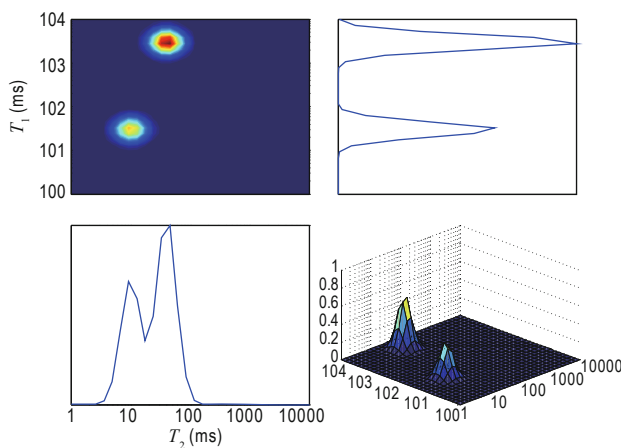
The improved truncated singular value decomposition (TSVD) algorithm is also used to solve most of the linear least squares problems. The TSVD method can give an optimal answer but loses the most powerful oscillation component of the solution, so the method reduces the results resolution which makes the spectrum discontinuous. In order to obtain a continuous spectrum, the solution is pointed to the minimum of $\|A\Delta x - \Delta b\|_2$ to solve for the optimal solution. First, assuming a known initial guess x_0 , the original equation (2) is rewritten as $A(x - x_0) = b - b_0$ or $A\Delta x = \Delta b$, which is executed by the TSVD algorithm. When $\|A\Delta x - \Delta b\|_2$ is a minimum, Δx is the optimal solution, so $x_0 + \Delta x$ is the final result of

equation (2). Since the equation solution of $A\Delta x = \Delta b$ is Δx , it does not reduce the matrix size in the non-negative constraints. If components of the solution are less than zero, these components are directly changed to zero and then the changed solutions are reiterated until the components meet the non-negative conditions. Because the singular value decomposition process of matrix $A_{m \times n}$ is executed only once, the amount of computations and computing time are greatly reduced.

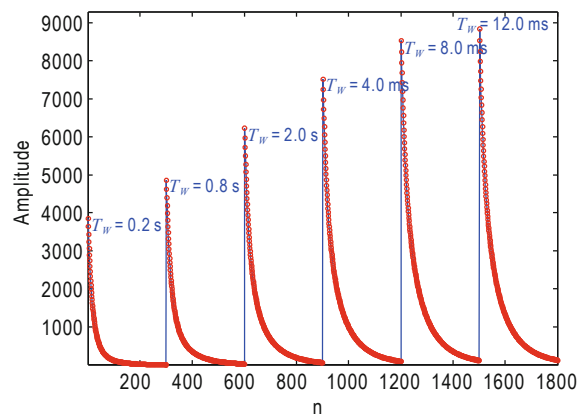
To verify the hybrid inversion method, a gas reservoir model containing bound water and free gas is built and the fluid properties are listed in Table 1. In the simulation experiment, the two-dimensional grid mesh with transverse relaxation time (T_2) and longitudinal relaxation time (T_1) is a 30×30 logarithmic uniform distribution (Tan et al., 2008). Fluid composition in the two-dimensional spectrum is a Gaussian distribution. The two-dimensional spectrum with multiple peaks is shown in Figure 1a. The fluid model observation parameters are: echo spacing (T_E) is 1.2 ms and the group of waiting times (T_w) is set to be 0.2s, 0.8s, 2.0s, 4.0s, 8.0s, and 12.0s. The number of echoes of each received echo train is 600. The magnetic field gradient is 2.0×10^{-3} T/cm and six groups of echo trains are simulated (see Figure 1b). In this case, the condition number of matrix $A_{3600 \times 900}$ is calculated to be about 4.0×1025 , which indicates that it is a serious pathological matrix. The LSQR and SVD algorithms are combined to solve the two-dimensional inversion problem.

Table 1 Reservoir fluid parameters

Fluid properties	T_1 (ms)	T_2 (ms)	Saturation (%)
Bound water	30	10	60.0
gas	3000	40	40.0



(a) Fluid model



(b) Synthesized echo trains

Fig.1 Fluid model and synthesized echo trains.

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Figures 2a and 2b illustrate the inversion results for single LSQR and SVD algorithms. The LSQR algorithm relative error is about 0.3046, whereas the SVD algorithm relative error is about 0.7054. We see that the LSQR algorithm inversion accuracy is higher than the SVD algorithm. The LSQR method running time is about 140.5 s and, by contrast, the SVD method running time with 800 iterations is about 1817.125 s. Therefore, the TSVD algorithm inversion speed is slower than the LSQR algorithm.

Figure 2a shows that the 2D spectrum inverted by

the LSQR algorithm is in very good agreement with the given model in Figure 1a. Furthermore, the inverted one-dimensional T_2 distribution (short T_2 components of 1 to 100 ms) is consistent with the one-dimensional T_1 distribution (long T_1 components of more than 100 ms) in the model. On the contrary, the inverted long T_2 and short T_1 components are not consistent with those of the supposed model. Therefore, the LSQR algorithm cannot achieve the qualitative analysis and quantitative calculations of reservoir fluid.

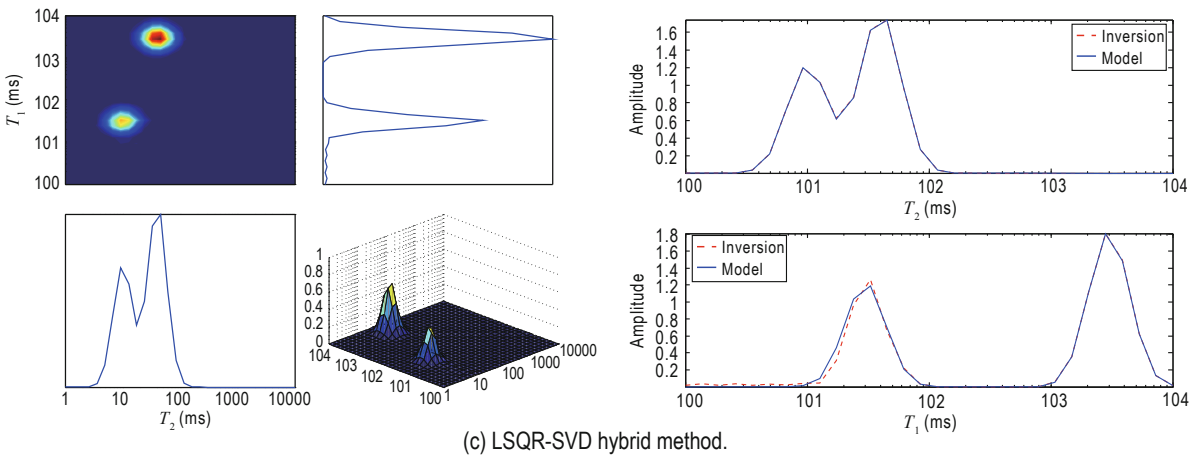
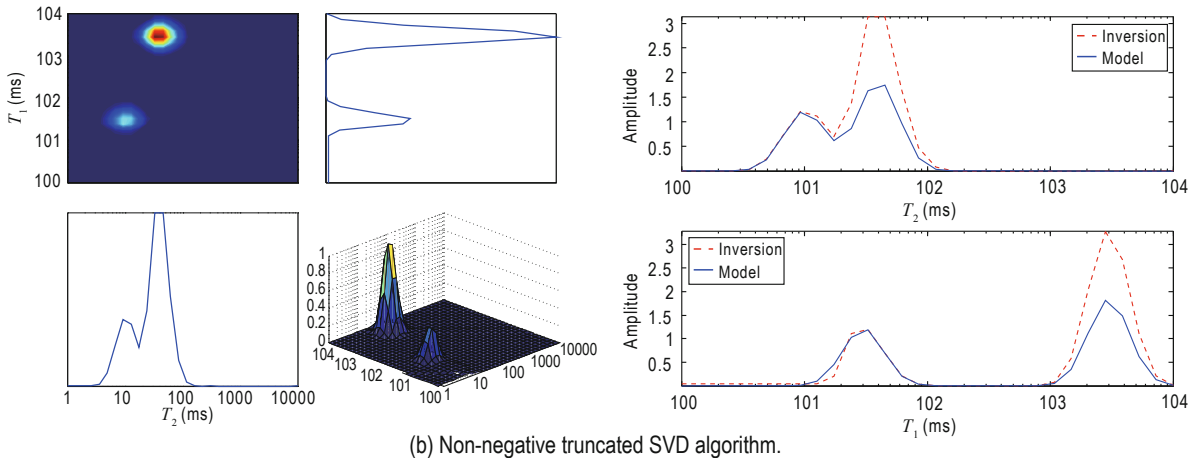
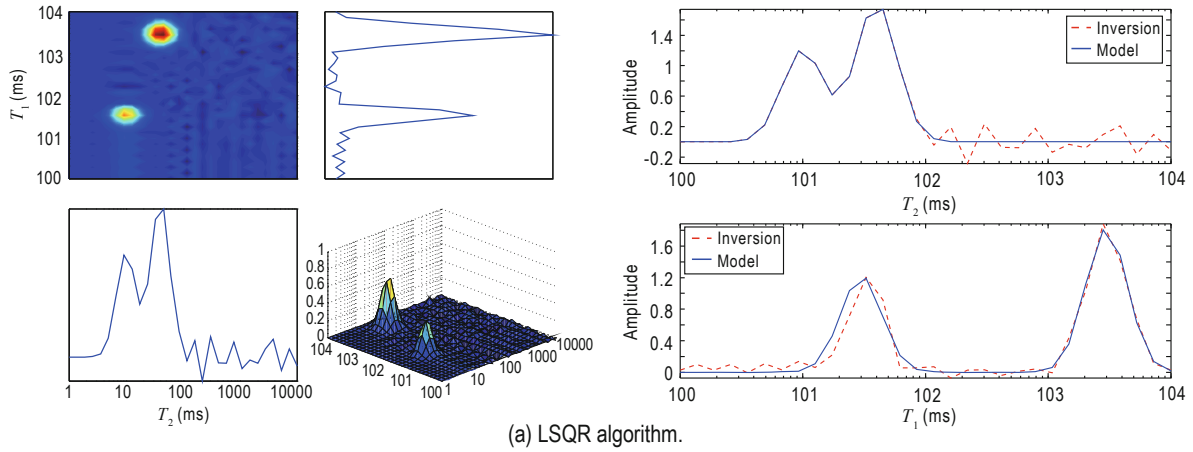


Fig.2 Comparison of different inversion algorithms for 2D NMR.

Similarly, Figure 2b shows two-dimensional inversion results of the TSVD algorithm with non-negative constraints. The two-dimensional spectrum looks like the given model in Figure 1a. The long one-dimensional T_2 distribution (>100 ms) relaxation components and short T_1 relaxation components of the one-dimensional T_1 distribution (<100 ms) match well with those of the given model in one-dimensional T_1 and T_2 distributions but the short T_2 components between 1 and 100 ms and long T_1 components between 100 and 3000 ms do not match with the given model. Comparisons of the two inversion algorithms indicate that the LSQR has faster inversion speed but lower accuracy. TSVD inversion accuracy can satisfy the requirements but the speed is relatively slow. Moreover, the TSVD algorithm solution is not complete due to decreasing the solution dimension and the lost components should be compensated for to improve the resolution and accuracy. From Figures 2a and 2b, the LSQR method for inversion of the short T_2 components is good, whereas the long T_2 components inverted by the improved TSVD method are better for inversion of long T_2 components. Therefore, a hybrid method with TSVD and LSQR algorithms is considered to elevate the inversion accuracy. That is, the inversion is first conducted by LSQR and then the results are set as initial values x_0 and then the TSVD inversion algorithm is used for further inversion.

Figure 2c shows the inversion results by the hybrid method, from which we see that the inversion results

are greatly improved. The T_2 inverted distribution is consistent with the model in short relaxation components as well as long components and the inverted T_1 distribution is also closer to the model. In the hybrid inversion process, the number of iterations for the LSQR method is about 800 and for the TSVD algorithm is 400 and the total computing time is 1722.41 s with inversion error of 0.07. Therefore, this hybrid method is more accurate and faster than a single LSQR or SVD inversion algorithm.

Observation parameter design and analysis

Observation parameter design is essential for fluid identification by nuclear magnetic resonance measurements. The data core matrix ($A_{m \times n}$) and the observed echo train (b_m) are not only related to the transverse and longitudinal relaxation time but also are affected by different combinations of waiting time and echo spacing (T_E). The influences of observation parameters on inversion results should be studied further. Therefore, the oil/water and gas/water models are designed for experimental study of inversion for distinguishing oil or gas from water. The fluid parameters are listed in Tables 2 and 3 and the corresponding models are shown in Figure 3.

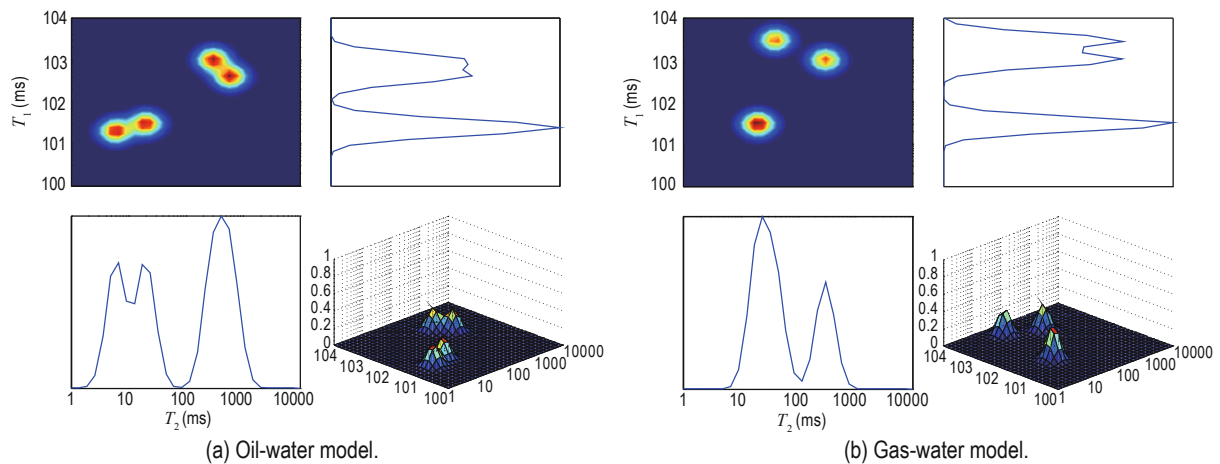


Fig.3 Fluid models based on (T_2 , T_1) 2D NMR.

Table 2 The properties of oil -water model

Fluids	Properties		
	T_1 (ms)	T_2 (ms)	Saturation (%)
Light oil	400	600	25.0
Viscous oil	20	6	25.0
Bound water	30	20	25.0
Free water	1000	300	25.0

Table 3 the properties of gas-water model

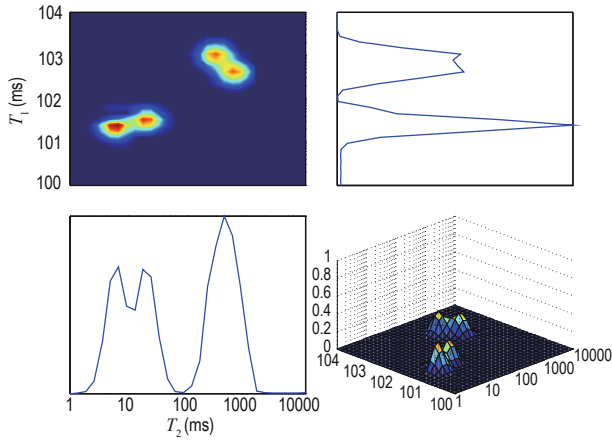
Fluids	Properties		
	T_1 (ms)	T_2 (ms)	Saturation (%)
Gas	3000	40	30.0
Bound water	30	20	40.0
Free water	1000	300	30.0

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During the experiment, the model is a 30×30 grid, echo spacing is 1.2 ms, the number of echoes is 600, and the magnetic field gradient is 3.0×10^{-3} T/cm with different waiting time (T_w) combinations and different echo spacing (T_E). First, six groups of echo trains are simulated and the hybrid method is used to invert the echo trains with different echo spacing or different combinations of waiting time. The inversion results show that the effect of different T_w combinations and T_E s on fluid identification are different (see Figure 3). The effect of different magnetic field gradients (G) in diverse fluid models need to be further studied. The oil, water, and gas contained in the formation are assumed to show a Gaussian distribution and multiple peak shapes. In forward and inversion simulations, transverse relaxation time (T_2) and longitude relaxation time (T_1) are divided into 30 bins in the range of 0.3 to 3000 ms and 10-power exponents are adopted (Tan et al, 2008). The gyromagnetic ratio (γ) is about $2\pi \times 42.58 \times 10^6$ Hz/T.

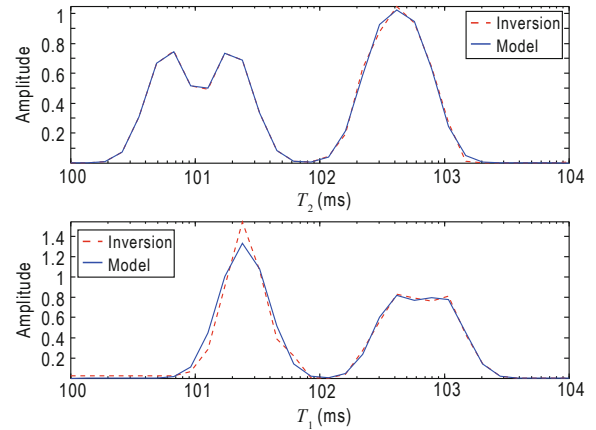
Different echo spacing (T_E)

For numerical simulations and inversion experiments of different echo spacing in the oil-water model, the observation parameters are: wait time (T_w) group is 0.05 s, 0.1 s, 1.0 s, 1.5 s, 3.0 s, 5.6 s, 8.0 s, 10 s, and 12 s and the number of echoes for each echo train is 300 for each waiting time. First, nine groups of echo trains are forward modeled with different echo spacing. Then, the hybrid inversion algorithm is used to invert the modeled data in order to study the effects of different echo spacing on the inversion results. Figure 4 shows the inversion results for different echo spacing in the oil-water model. When the echo spacing is 0.6 ms, the relative error between the inversion results and the model is about 0.1678. When the echo spacing is 1.2 ms, the relative error is about 0.1614. When the echo spacing is 2.4 ms, the relative error is about 0.1572. From Figure 4, we see that the echo spacing has almost no impact on the accuracy of the inversion for the oil-water model.

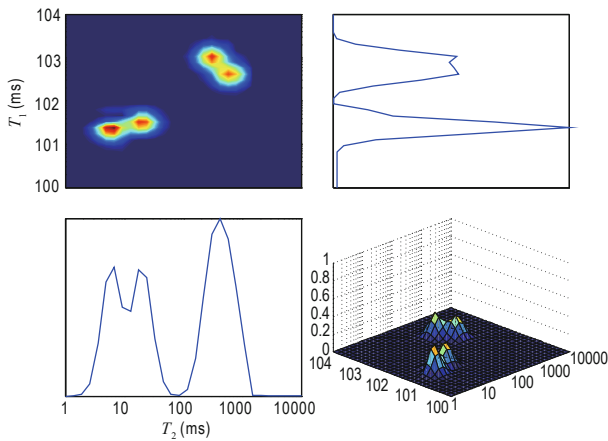


Two-dimensional spectrum

(a) $T_E = 0.6$ ms

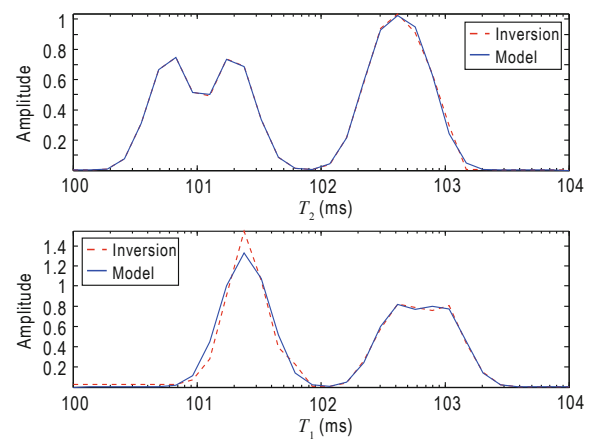


Inversion results comparison



Two-dimensional spectrum

(b) $T_E = 1.2$ ms



Inversion results comparison

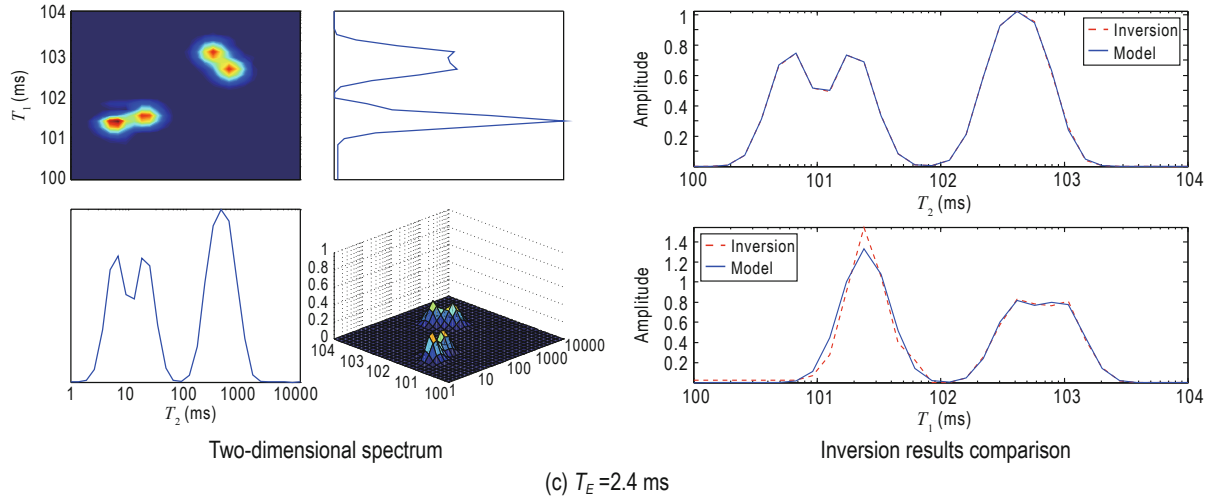
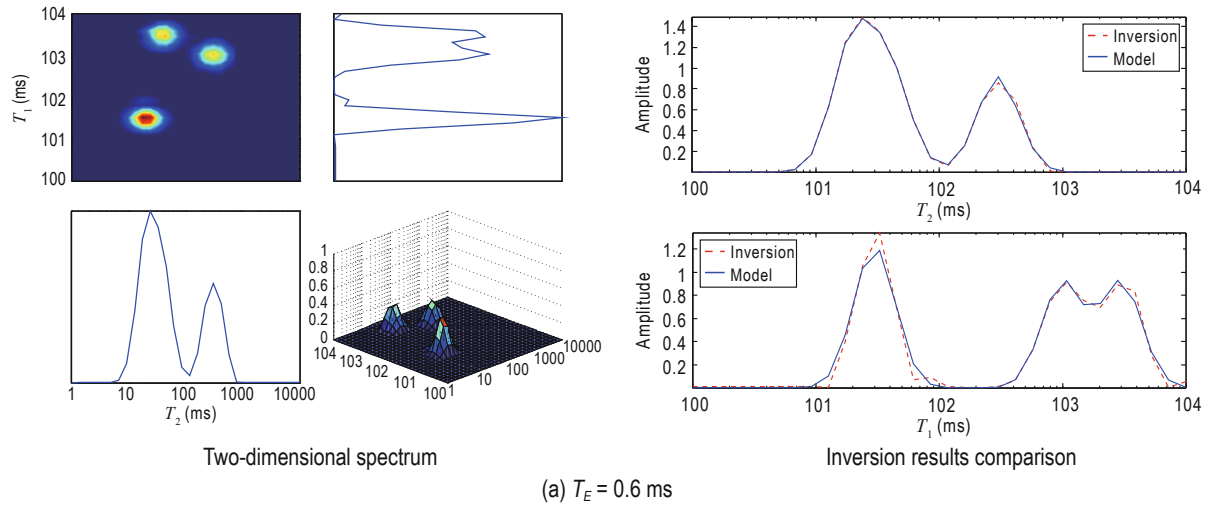


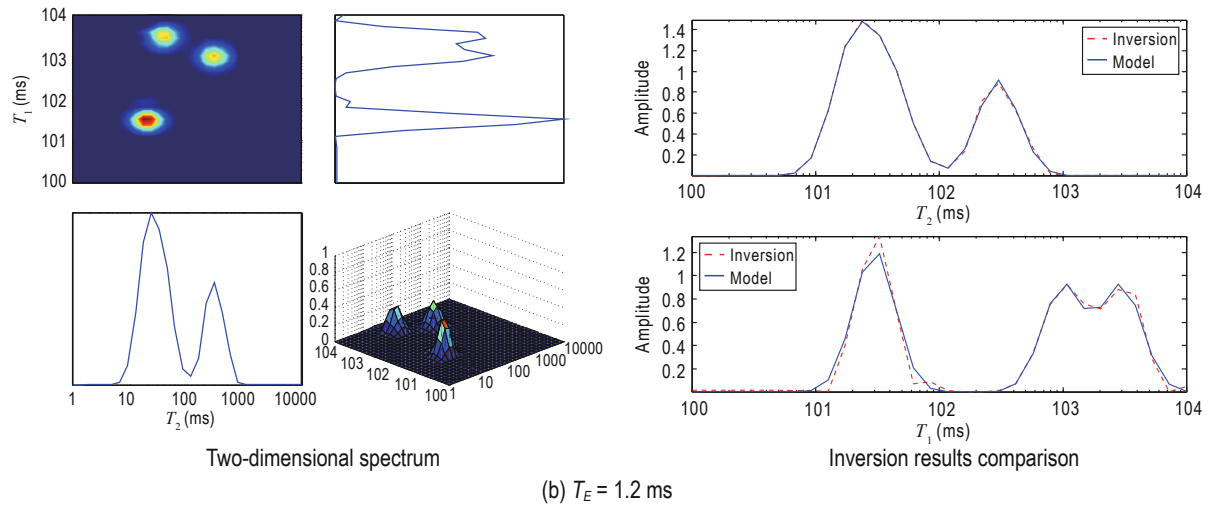
Fig.4 Comparisons of 2D NMR inversion results with different T_E s for the oil-water model.

Similarly, for the gas-water model, a similar experiment was carried out. When the echo spacing is 0.6 ms, the relative error between the inversion results and gas-water model is about 0.1199. When the echo spacing is 2.4 ms,

the inversion relative error (R_x) is about 0.1155. From the simulation results we see that in the multiple T_w observation modes, the echo spacing has almost no impact on the inversion accuracy for the gas-water and oil-water models.



(a) $T_E = 0.6$ ms



(b) $T_E = 1.2$ ms

(T_2, T_1) 2D NMR and fluid responses

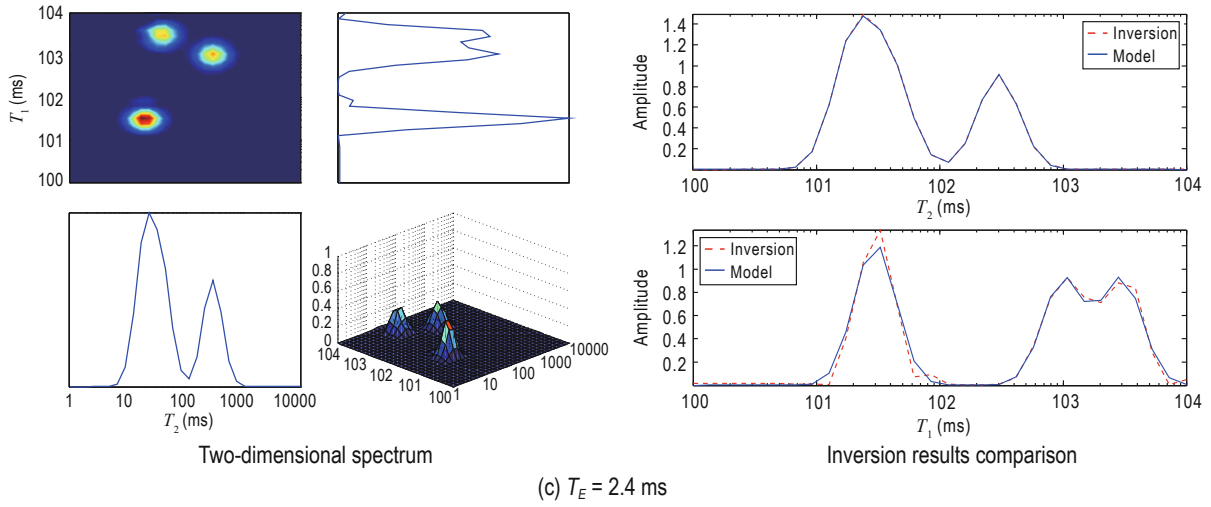


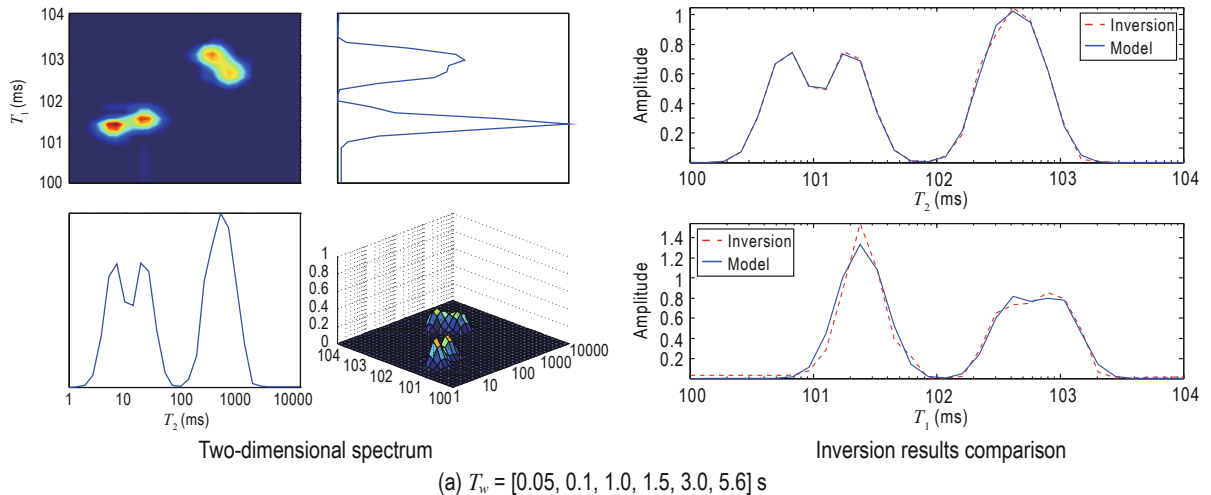
Fig.5 Comparisons of 2D NMR inversion results with different T_E s for the gas-water model.

Different waiting time (T_w) groups

For studying the influences of waiting times (T_w) on the inversion results, different waiting time groups are set for a series of simulation and inversion experiments. The observation parameters are: the echo spacing is 0.6 ms, the number of echoes is 300 for each echo train corresponding to each waiting time. The waiting time groups are set to 0.05 s, 0.1 s, 1.0 s, 1.5 s, 3.0 s, 5.6 s, 6.0 s, 6.5 s, 7.0 s, 8.0 s, 10.0 s, and 12.0 s and 0.05 s, 0.1 s, 1.0 s, 3.0 s, 8.0 s, and 12.0 s. First, six groups of echo trains are simulated and then the simulation data is inverted using the hybrid inversion method. The inversion results of different fluid models are illustrated in Figures 6 and 7.

When the waiting time group is chosen as 0.05 s, 0.1 s, 1.0 s, 1.5 s, 3.0 s, and 5.6s, the relative error (R_x) is about 0.1877. When waiting time group is 6.0 s, 6.5 s, 7.0 s, 8.0 s, 10.0 s, and 12.0s, the relative error (R_x) is

about 0.7559. When the waiting time group is 0.05 s, 0.1 s, 1.0 s, 3.0 s, 8.0 s, and 12.0 s, the relative error (R_x) is about 0.2031. From 2D spectrum inversion and error analysis, the results with the waiting time group of 0.05 s, 0.1 s, 1.0 s, 3.0 s, 8.0 s, and 12.0s is best in three tests in which every fluid peak is focused well and the T_2 and T_1 distributions also match well with the model. Therefore, in multiple T_w observation modes, for the oil-water model, the inversion results of the mixed waiting time group with short and long T_w not only reflect the bound water with short T_2 relaxation time and short T_1 relaxation time but can also identify the components with long T_2 and T_1 relaxation times in porous media. From two-dimensional inversion spectra, light oil and viscous oil also easily can be distinguished from movable water and bound water.



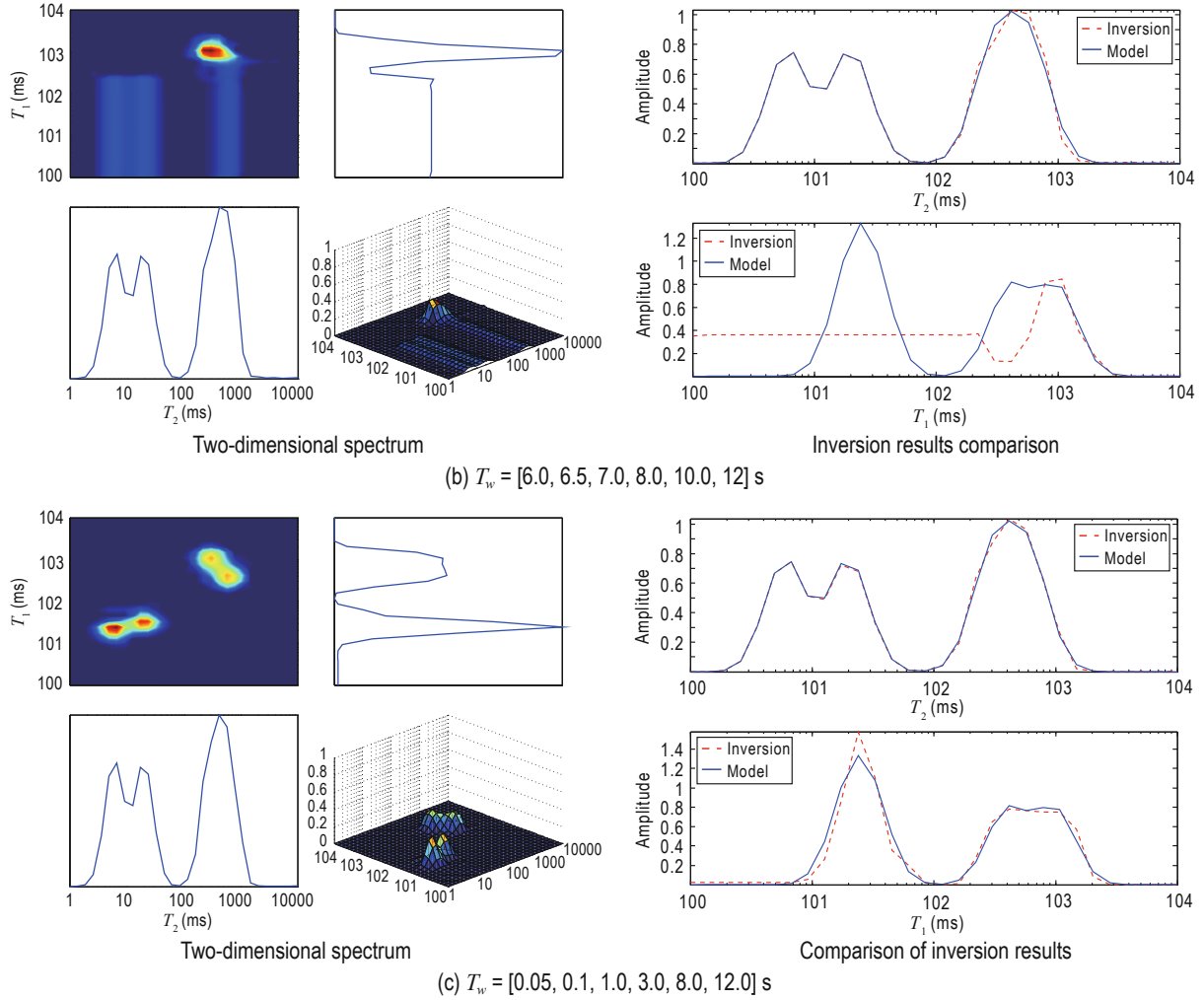
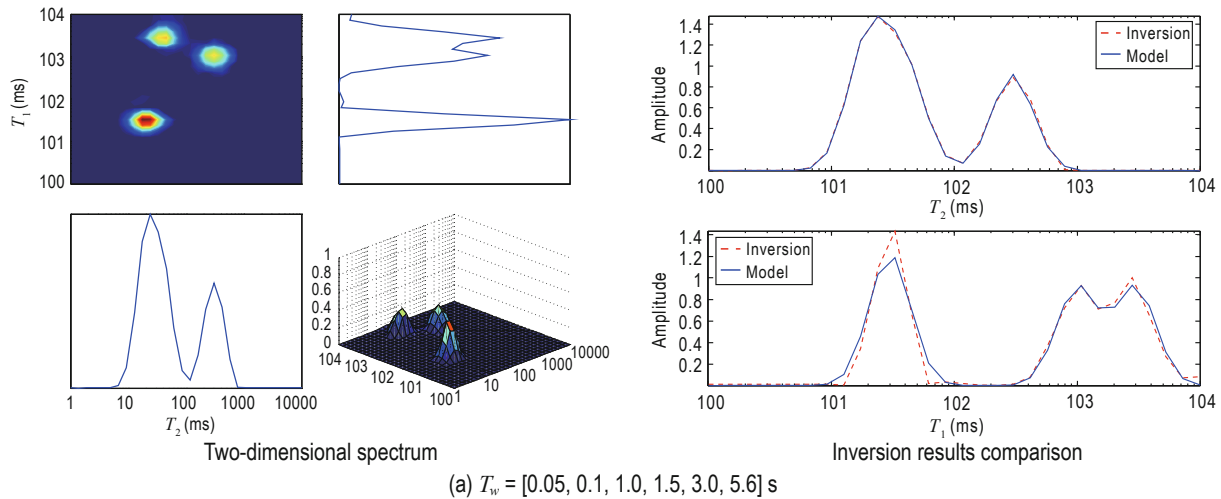


Fig.6 Comparisons of 2D NMR inversion results with different T_w groups for the oil-water model.

A series of similar experiments was made for the gas-water model using a group of waiting times of 0.05 s, 0.5 s, 0.1 s, 1.0 s, 1.5 s, 3.0 s, and 5.6 s. The experiments show that the relative error (R_x) between the model and inversion results is about 0.1885. When the group of

waiting times is 6.0 s, 6.5 s, 7.0 s, 8.0 s, 10.0 s, and 12.0 s, the relative error (R_x) is about 0.6220. When the group of waiting time is 0.05 s, 0.1 s, 1.0 s, 3.0 s, 8.0 s, and 12.0s, the relative error (R_x) between model and inversion is about 0.1923.



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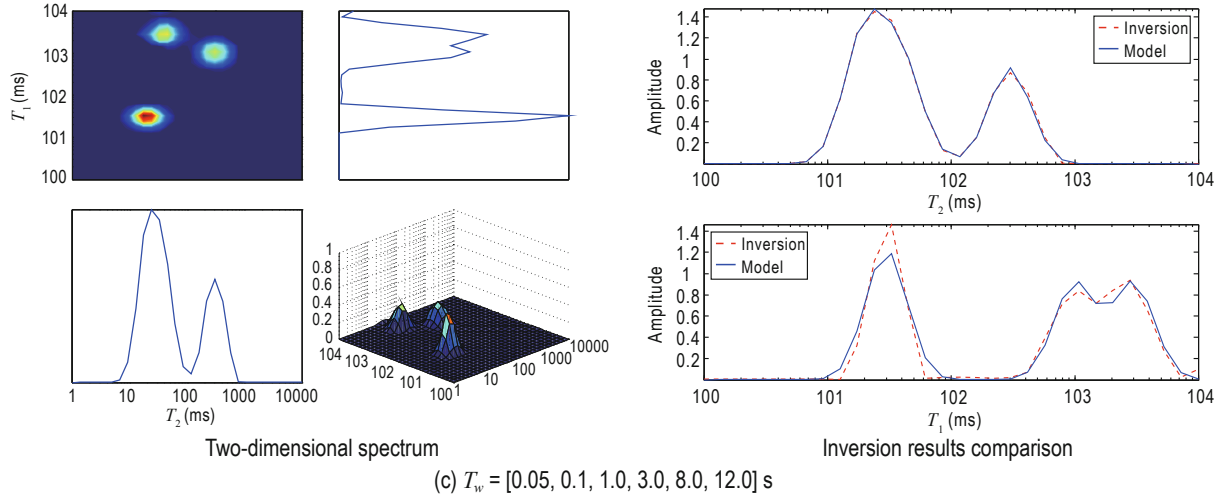
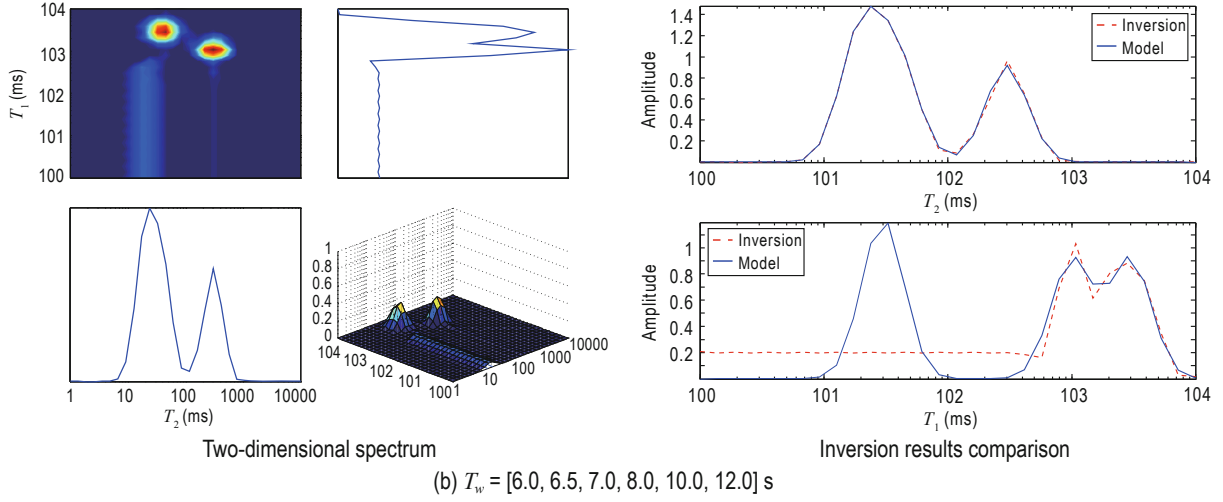


Fig.7 Comparisons of 2D NMR inversion results with different T_w groups for the gas-water model.

From the inversion results and error analysis, we see that for both fluid models, the error of inversion results with mixed- T_w groups of short waiting time T_w and long waiting time T_w is smallest in three single T_w groups. So, it is most suitable for (T_2, T_1) 2D NMR logging.

Effects of SNRs

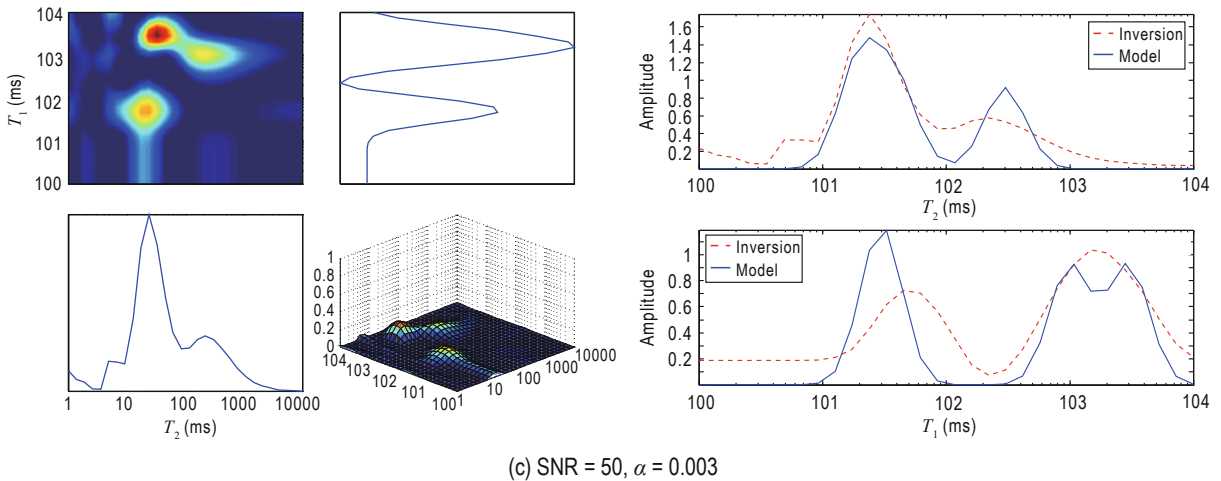
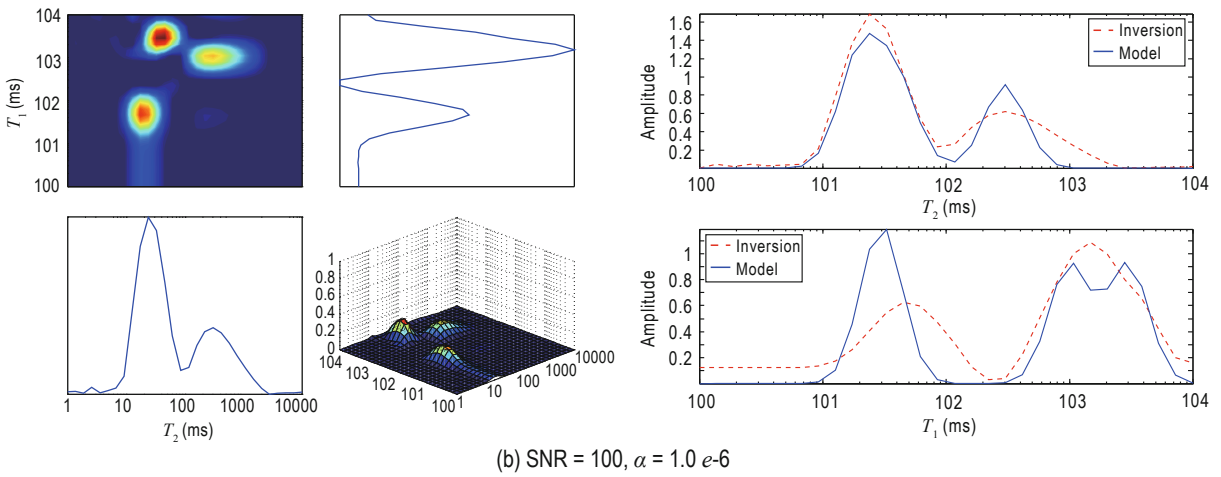
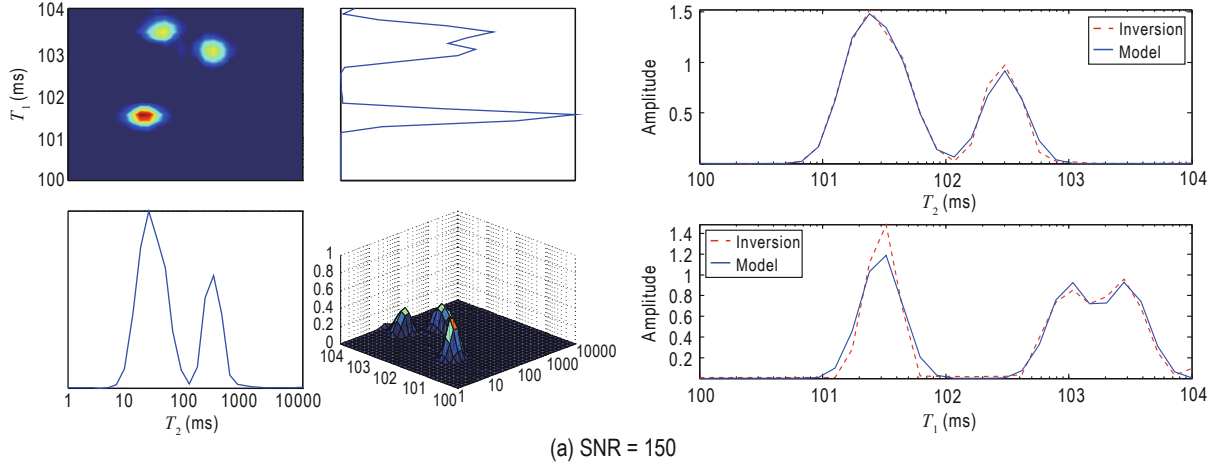
For the gas-water model, we chose different single-to-noise-ratios (SNR ratios) to study the effects of different algorithms on (T_2, T_1) 2D NMR inversion results. In simulation experiments, the echo spacing (T_E) is set to 0.6 ms, the multiple T_w group is [0.05, 0.1, 1.0, 3.0, 8.0, 12] s. The different SNR ratios are added into the forward echo trains and the inversions are carried out by the hybrid inversion method.

When the SNR ratio is 150, the inversion results are illustrated in Figure 8a and the relative error (R_x) is about 0.2113. The gas can be distinguished from water in the two-dimensional spectrum and one-dimensional T_2 and D distributions shown in Figure 8. The inversion process may be directly executed by the hybrid inversion method when the SNR ratio is greater than 150.

When the SNR ratio is less than 150, the damping factor (α) should be added into the hybrid inversion. The inversion problem of $Ax = b$ is changed to the solution of $(A^T A + \alpha I)x = A^T b$, then the hybrid inversion method is used. Different SNR ratios such as 100, 50, and 25 were chosen for numerical simulation experiments. The inversion results are shown in Figures 8b, 8c, and 8d and the relative errors are about 0.5472, 0.6594, and 0.7968, respectively. From Figures 8b and 8c, we see that when the SNR ratio is 100 or 50, both the two-dimensional spectrum and the one-dimensional

T_2 distribution and diffusion coefficients can directly be used to identify the gas and water. When the SNR ratio is lower than 50, the gas and water are mixed (see Figure 8d) and it is difficult to distinguish between

them. The SNR ratio is lower and the fluid identification is more difficult for the gas-water model. However, when the SNR ratio is greater than 50, (T_2 , T_1) 2D NMR can identify fluids more accurately.



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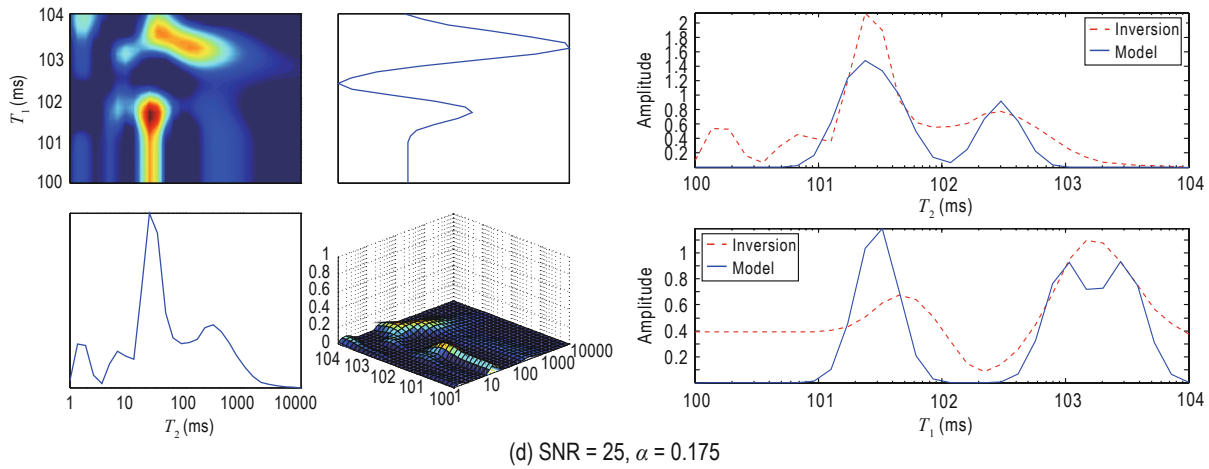


Fig.8 Inversion results with different SNRs for gas-water models.

The experiments were also performed for the oil-water models. The experiment results show that when the SNR ratio is 150, heavy oil can be distinguished from bound water in the viscous oil and water model but the movable water and light oil overlap each other in the light oil-water model, so they are difficult to be identified. When the SNR is 100, the viscous oil and bound water can still be distinguished from each other in the heavy oil-water model. However, the movable water and light oil overlap and light oil is not distinguished from movable water in the light oil and water model. When the SNR ratio is lower than 50, viscous oil and bound water overlap and movable water and light oil also overlap. From the experiments, we conclude that the SNR ratio effect on inversion results in the gas-water model is lower than in the oil-water model for fluid identification.

Conclusions

The proposed non-negative LSQR and SVD hybrid inversion method achieves (T_2, T_1) 2D NMR inversion with multiple- T_w activation. The hybrid method can be used through redesigned acquisition parameters. Numerical simulation experiments show that the hybrid inversion method can meet the needs of practical application. From the inversion results and error analysis, we conclude that:

(1) The hybrid inversion method is stable and more accurate than a single inversion algorithm which decomposes a matrix only once. It shows fast calculation speed and only needs small storage space. The algorithm easily achieves the non-negative constraints and can get a continuous 2D high-precision spectrum.

(2) With respect to the acquisition parameters, the mixed waiting time combination with short and long waiting times are more suitable for fluid identification. The results identify bound fluids with the fast relaxation components and oil or gas and water with the slow relaxation component.

(3) The SNR ratio effect on inversion results in the gas-water model is lower than in the oil-water model for fluid identification.

(4) The reservoir fluids can be qualitatively determined by the location and amplitude of the focused signals in the two-dimensional spectrum. For light oil-water and gas-water models, both oil and gas are distinguished from water more easily, whereas for fluid models containing viscous oil, the oil and water are not easily identified because the NMR relaxation properties of bound fluids and viscous oil are very similar.

References

- Bi, L. R., 2007, Latest advances in NMR logging technology: Chinese Journal of Engineering Geophysics, **4**(4), 369 – 374.
- Coates, G. R., Xiao, L. Z., and Prammer, M. G., 1998, NMR logging principle & applications: Gulf Publishing Company, Texas.
- Dunn, K. J., and Latorraca, G. A., 1999, The inversion of NMR log data sets with different measurement errors: Journal of Magnetic Resonance, **140**, 153 – 161.
- Gu, Z. B., and Liu, W., 2007, The inversion of two-dimensional NMR map: Chinese Journal of Magnetic Resonance, **24**(3), 311 – 319.
- Gu, Z. B., Liu, W., and Sun, D. Q., 2009, Application of 2D

- NMR techniques in petroleum logging: Chinese Journal of Magnetic Resonance, **26**(4), 560 – 568.
- Hurlimann, M. D., Venkataramanan, L., and Flaum, C., et al., 2002, Diffusion-editing: New NMR measurement of saturation and pore geometry: Paper FFF in 43rd Annual Symposium of SPWLA, Oiso, Japan, June 3 – 6.
- Liu, J. S., and Liu, F. T., 2006, Seismic tomography LSQR algorithm parallelization: Geophysics, **49**(2), 540 – 545.
- Shao, W. Z., 2003, On the effect of NMR differential and shifted spectrum method in laboratory: Well Logging Technology, **27**(6), 502 – 507.
- Sun, B. Q., and Dunn, K. J., 2005, A global inversion method for multi-dimensional NMR logging: Journal of Magnetic Resonance, **40**(2), 152 – 160.
- Tan, M. J., Zhao, W. J., and Fan, Y. R., 2006, Identification of reservoir fluid with double- T_w NMR logging activity properties: Natural Gas Industry, **26**(4), 38 – 40.
- Tan, M. J., Shi, Y. L., and Zhao, W. J., et al., 2008, A joint inversion method for NMR dual- T_w logging data and fluid typing: Chinese Journal of Geophysics, **51**(5), 1100 – 1109.
- Tan, M. J., Zou, Y. L., and Liu, B. K., et al., 2011, Inversion simulation of (T_2, D) 2D NMR logging and analysis of observation parameters effects in gas-water model: Well Logging Technology, **35**(2), 130 – 136.
- Wang, Z. D., Wang, H., and Xiang, T. D., 2001, Integration of NMR DT_w logs and DT_E logs to improve the oil bed interpretation accuracy: Well Logging Technology, **25**(5), 365 – 368.
- Xiao, L. Z., 1998, Nuclear magnetic imaging logging, rock NMR experiment and applications: Science Press, Beijing.
- Xiao, L. Z., 2007, Some important issues for NMR logging applications in China: Well Logging Technology, **31**(5), 401 – 407.
- Xie, R. H., Xiao, L. Z., Dunn, K. J., et al., 2005, Two-dimensional NMR logging: Well Logging Technology, **29**(5), 430 – 434.
- Xie, R. H., Xiao, L. Z., and Fu, S. Q., et al., 2007, Experimental study on nuclear magnetic resonance relaxation characteristic of crude oil at variable temperature: Journal of China University of Petroleum, **31**(4), 34 – 37.
- Xie, R. H., Xiao, L. Z., and Lu, D. W., 2009, (T_2, T_1) Two-dimensional NMR method for fluid typing: Well Logging Technology, **33**(1), 10 – 20.
- Xie, R. H., and Xiao, L. Z., 2009, Two-dimensional NMR logging method for identifying reservoir fluids: Geophysics, **52**(9), 2410 – 2418.
- Yang, W., Liu, S. X., and Feng, Y. Q., 2008, Cross-hole tomography LSQR algorithm: Geophysical and Geochemical Exploration, **32**(2), 199 – 202.
- Yun, H. Y., and Tan, M. J., 2006, Observation way and analytical approach of dual waiting time by nuclear magnetic resonance logging: Petroleum Geology and Recover Efficiency, **13**(4), 96 – 98.

Tan Mao-Jin, an Associate Professor at China University of Geosciences (Beijing). He gained a PhD at China University of Petroleum in 2006 and was a postdoctoral fellow of GUCAS in Geophysics from 2006 to 2008. He is interested in well logging theory and logging interpretation and evaluation of complex reservoirs.